

Reinforcement Learning Theory

CS 185/285

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Part 1:

What's the point of RL theory?

What questions do we ask in RL theory?

Lots of different questions! But here are a few common ones:

If I use this algorithm with N samples, k iterations, how good is the result?

Let's say we're doing Q-learning...

$$\|\hat{Q}_k - Q^*\| \leq \epsilon \quad \text{with probability at least } 1 - \delta \quad \text{if } N \geq f(\epsilon, \delta)$$

$$\|Q^{\pi_k} - Q^*\| \leq \epsilon$$

We'll focus on these types of questions today

↙ not the same thing! Q^{π_k} is the *true* Q -function of policy at iteration k

If I use this exploration algorithm, how high is my regret?

$$\text{Reg}(T) \leq \mathcal{O} \left(\sqrt{T \cdot N \cdot \log \frac{NT}{\delta}} \right) + \delta T$$

But there are many others!

What kinds of assumptions do we make?

Effective analysis is **very** hard in RL without strong assumptions

The trick is to make assumptions that admit interesting conclusions without divorcing us (too much) from reality

Exploration: Performance of RL methods is greatly complicated by exploration – how likely are we to find (potentially sparse) rewards?

Theoretical guarantees typically address worst case performance, and worst case exploration is extremely hard

Goal: show that exploration method (e.g., counts) is $\text{Poly}(|S|, |A|, H)$

Learning: If we somehow “abstract away” exploration, how many samples do we need to effectively learn a model or value function that results in good performance?

“generative model” assumption: assume we can sample from $P(s'|s, a)$ for *any* (s, a)

“oracle exploration”: for every (s, a) , sample $s' \sim P(s'|s, a)$ N times

What's the **point**?

1. Prove that our RL algorithms will work perfectly every time

Usually not possible with current deep RL methods, which are often not even guaranteed to converge

2. Understand how errors are affected by problem parameters

Do larger discounts work better than smaller ones?

If we want half the error, do we need 2x the samples? 4x? something else?

Usually we use precise theory to get imprecise **qualitative** conclusions about how various factors influence the performance of RL algorithms under **strong assumptions**, and try to make the assumptions reasonable enough that these conclusions are likely to apply to real problems (but they are not guaranteed to apply to real problems)

Don't take someone seriously if they say their RL algorithm has "provable guarantees" – the assumptions are always unrealistic, and theory is at best a rough guide to what might happen

Some useful identities

$$Q^\pi(s, a) = r(s, a) + \gamma E_{s' \sim P(s'|s, a)}[V^\pi(s')] = r(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s')$$

$$Q^\pi = r + \gamma P V^\pi$$

Diagram illustrating the dimensions of the matrices and vectors in the equation $Q^\pi = r + \gamma P V^\pi$:

- Q^π : A vertical blue bar representing a vector of size $|S||A|$.
- r : A vertical blue bar representing a vector of size $|S||A|$.
- P : A 3x3 blue grid representing a matrix of size $|S||A| \times |S|$.
- V^π : A vertical blue bar representing a vector of size $|S|$.

$$P^\pi(s', a'|s, a) = \pi(a'|s')P(s'|s, a)$$

how big is this?

say $r \approx 1$

$$\sum_{t=0}^{\infty} \gamma^t r \approx \sum_{t=0}^{\infty} \gamma^t = \frac{1}{1-\gamma}$$

“horizon-like term”

$$Q^\pi = r + \gamma P^\pi Q^\pi$$

$$Q^\pi - \gamma P^\pi Q^\pi = r$$

$$(\mathbf{I} - \gamma P^\pi) Q^\pi = r$$

$$Q^\pi = (\mathbf{I} - \gamma P^\pi)^{-1} r$$

$$\sum_{t=1}^H r \approx \sum_{t=1}^H 1 = H$$

$$P^\pi$$

Diagram illustrating the dimensions of the matrix P^π :

- P^π : A 4x4 blue grid representing a matrix of size $|S||A| \times |S||A|$.

policy evaluation is a linear operation

Part 2:

A warmup: convergence of VI

Let's prove that value iteration converges

value iteration: $V \leftarrow \max_a [r + \gamma PV]$

$$TV = \max_a [r + \gamma PV]$$

“Bellman optimality operator”

$$|\max_x f(x) - \max_x g(x)| \leq \max_x |f(x) - g(x)|$$

$$\begin{aligned} \|TV - TU\|_\infty &\leq |TV(s) - TU(s)| = |\max_a [r(s, a) + \gamma \sum_{s'} P(s'|s, a)V(s')] - \max_a [r(s, a) + \gamma \sum_{s'} P(s'|s, a)U(s')]| \\ &\leq \max_a |r(s, a) + \gamma \sum_{s'} P(s'|s, a)V(s') - r(s, a) - \gamma \sum_{s'} P(s'|s, a)U(s')| \\ &= \gamma \max_a |\sum_{s'} P(s'|s, a)(V(s') - U(s'))| \\ &\leq \gamma \max_{s'} |V(s') - U(s')| \\ &= \gamma \|V - U\|_\infty \end{aligned}$$

Let's prove that value iteration converges

value iteration: $V \leftarrow \max_a [r + \gamma PV]$

$$TV = \max_a [r + \gamma PV] \quad \text{“Bellman optimality operator”}$$

$\|TV - TU\|_\infty \leq \gamma \|V - U\|_\infty$ applying T to any two value functions V and U brings them closer

$$TV^* = V^*$$

$\|TV - V^*\|_\infty = \|TV - TV^*\|_\infty \leq \gamma \|V - V^*\|_\infty$ applying T to V brings it closer to V^*

$$\|T^k V - V^*\|_\infty \leq \gamma^k \|V - V^*\|_\infty$$

$$\lim_{k \rightarrow \infty} \|T^k V - V^*\|_\infty \rightarrow 0$$

Intermission

Part 3:

Sample complexity without exploration

The setup

“oracle exploration”: for every (s, a) , sample $s' \sim P(s'|s, a)$ N times

simple “model based” algorithm:

1. $\hat{P}(s'|s, a) = \frac{\#(s, a, s')}{N}$
2. Given π , use $\hat{P}$ to estimate $\hat{Q}^\pi$

how close is $\hat{Q}^\pi$ to Q^π ? $\longleftarrow$

$$\|Q^\pi(s, a) - \hat{Q}^\pi(s, a)\|_\infty \leq \epsilon \quad \begin{array}{l} \text{with probability at least } 1 - \delta \\ \text{if } N \geq f(\epsilon, \delta) \end{array}$$

$$\max_{s, a} |Q^\pi(s, a) - \hat{Q}^\pi(s, a)| \leq \epsilon$$

good to use $\|\cdot\|_\infty$ if we want *worst-case* performance

how close is $\hat{Q}^*$ if we learn it using $\hat{P}$? $\longleftarrow$

$$\|Q^*(s, a) - \hat{Q}^*(s, a)\|_\infty \leq \epsilon$$

optimal Q-function learned under $\hat{P}$

how good is the resulting policy? $\longleftarrow$

$$\|Q^*(s, a) - Q^{\hat{\pi}}(s, a)\|_\infty \leq \epsilon$$

the arg max policy corresponding to that Q-function

Concentration inequalities

whenever we need to answer questions about how close a learned function is to the true function, in terms of # of samples

Lemma A.1. (Hoeffding's inequality) Suppose $X_1, X_2, \dots, X_n$ are a sequence of independent, identically distributed (i.i.d.) random variables with mean μ . Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Suppose that $X_i \in [b_-, b_+]$ with probability 1, then

$$P(\bar{X}_n \geq \mu + \epsilon) \leq e^{-2n\epsilon^2/(b_+ - b_-)^2}.$$

Similarly,

$$P(\bar{X}_n \leq \mu - \epsilon) \leq e^{-2n\epsilon^2/(b_+ - b_-)^2}.$$

interpretation:

if we estimate μ with n samples the probability we're off by more than ϵ is at most $2e^{-2n\epsilon^2/(b_+ - b_-)^2}$

equivalently, if we want this probability to be δ :

$$\delta \leq 2e^{-2n\epsilon^2/(b_+ - b_-)^2} \Rightarrow \log \frac{\delta}{2} \leq -2n\epsilon^2/(b_+ - b_-)^2 \Rightarrow \frac{(b_+ - b_-)^2}{2n} \log \frac{2}{\delta} \geq \epsilon^2 \Rightarrow \frac{b_+ - b_-}{\sqrt{2n}} \sqrt{\log \frac{2}{\delta}} \geq \epsilon$$

$$\text{or... } n \leq \frac{(b_+ - b_-)^2}{2\epsilon^2} \log \frac{2}{\delta}$$

error (ϵ) scales as $\frac{1}{\sqrt{n}}$

Concentration inequalities

$$\hat{P}(s'|s, a) = \frac{\#(s, a, s')}{N} \quad \text{discrete distribution}$$

Proposition A.8. (*Concentration for Discrete Distributions*) Let z be a discrete random variable that takes values in $\{1, \dots, d\}$, distributed according to q . We write q as a vector where $\vec{q} = [\Pr(z = j)]_{j=1}^d$. Assume we have N iid samples, and that our empirical estimate of $\vec{q}$ is $[\hat{q}]_j = \sum_{i=1}^N \mathbf{1}[z_i = j]/N$.

We have that $\forall \epsilon > 0$:

$$\Pr \left(\|\hat{q} - \vec{q}\|_2 \geq 1/\sqrt{N} + \epsilon \right) \leq e^{-N\epsilon^2}.$$

which implies that:

$$\Pr \left(\|\hat{q} - \vec{q}\|_1 \geq \sqrt{d}(1/\sqrt{N} + \epsilon) \right) \leq e^{-N\epsilon^2}.$$

$$\delta \leq e^{-N\epsilon^2} \Rightarrow \epsilon \leq \frac{1}{\sqrt{N}} \sqrt{\log \frac{1}{\delta}} \quad \|\hat{P}(s'|s, a) - P(s'|s, a)\|_1 \leq \sqrt{|S|}(1/\sqrt{N} + \epsilon) \quad \text{with prob } 1 - \delta$$

$$\Rightarrow N \leq \frac{1}{\epsilon^2} \log \frac{1}{\delta} \quad \|\hat{P}(s'|s, a) - P(s'|s, a)\|_1 \leq \sqrt{\frac{|S|}{N}} + \sqrt{\frac{|S| \log 1/\delta}{N}} \leq c \sqrt{\frac{|S| \log 1/\delta}{N}}$$

A few useful lemmas

Next goal: relate error in $\hat{P}$ to error in $\hat{Q}^\pi$

Relating P to Q^π :

$$Q^\pi(s, a) = r(s, a) + \gamma E_{s' \sim P(s'|s, a)}[V^\pi(s')]$$

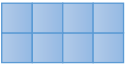
$$Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s')$$

$$Q^\pi = r + \gamma P^\pi Q^\pi$$

$$Q^\pi - \gamma P^\pi Q^\pi = r$$

$$(I - \gamma P^\pi) Q^\pi = r$$

$$Q^\pi = (I - \gamma P^\pi)^{-1} r$$

	Q^π	r	P	V^π
$Q^\pi = r + \gamma P V^\pi$	 $ S A $	 $ S A $	 $ S A $	 $ S $
			$ S $	
$V^\pi = \Pi Q^\pi$	Π			
	 $ S $			
	$ S A $			
			$Q^\pi = r + \gamma P^\pi Q^\pi$	
			$P^\pi = P\Pi$	

A few useful lemmas

$$Q^\pi = (I - \gamma P^\pi)^{-1} r \quad \hat{Q}^\pi = (I - \gamma \hat{P}^\pi)^{-1} r$$

Simulation lemma:

$$Q^\pi - \hat{Q}^\pi = \underbrace{\gamma(I - \gamma \hat{P}^\pi)^{-1}}_{\text{evaluation operator}} \underbrace{(P - \hat{P})}_{\text{difference in probabilities}} V^\pi$$

true value 

$$\begin{aligned} Q^\pi - \hat{Q}^\pi &= Q^\pi - (I - \gamma \hat{P}^\pi)^{-1} r \\ &= (I - \gamma \hat{P}^\pi)^{-1} (I - \gamma \hat{P}^\pi) Q^\pi - (I - \gamma \hat{P}^\pi)^{-1} r \\ &= (I - \gamma \hat{P}^\pi)^{-1} (I - \gamma \hat{P}^\pi) Q^\pi - (I - \gamma \hat{P}^\pi)^{-1} (I - \gamma P^\pi) Q^\pi \\ &= (I - \gamma \hat{P}^\pi)^{-1} ((\cancel{I} - \gamma \hat{P}^\pi) - (\cancel{I} - \gamma P^\pi)) Q^\pi \\ &= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P^\pi - \hat{P}^\pi) Q^\pi \\ &= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P\Pi - \hat{P}\Pi) Q^\pi \\ &= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) \Pi Q^\pi \\ &= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) V^\pi \end{aligned}$$

A few useful lemmas

Another useful lemma: given P^π and *any* vector $v \in \mathbb{R}^{|S||A|}$, we have:

$$\|(I - \gamma P^\pi)^{-1}v\|_\infty \leq \|v\|_\infty / (1 - \gamma)$$

“Q-function” corresponding to “reward” v is at most $1/(1 - \gamma)$ times larger

let $w = (I - \gamma P^\pi)^{-1}v$

$$\sum_{t=0}^{\infty} \gamma^t c = \frac{c}{1 - \gamma}$$

↑

$$\|v\|_\infty = \|(I - \gamma P^\pi)w\|_\infty \geq \|w\|_\infty - \gamma \|P^\pi w\|_\infty \geq \|w\|_\infty - \gamma \|w\|_\infty = (1 - \gamma) \|w\|_\infty$$

↑
triangle inequality

$$\|a - b\| \geq \|a\| - \|b\|$$

↑

$$\|P^\pi\|_\infty \leq 1$$

$$\|v\|_\infty / (1 - \gamma) \geq \|w\|_\infty$$

Putting them together...

$$\|(I - \gamma P^\pi)^{-1} v\|_\infty \leq \|v\|_\infty / (1 - \gamma)$$

$$Q^\pi - \hat{Q}^\pi = \gamma(I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) V^\pi$$

can we bound $\|V^\pi\|_\infty$? assume $R_{\max} = 1$

$$\sum_{t=0}^{\infty} \gamma^t r_t \leq \sum_{t=0}^{\infty} \gamma^t R_{\max} = \frac{1}{1 - \gamma} R_{\max}$$

$$\|\hat{P}(s'|s, a) - P(s'|s, a)\|_1 \leq c \sqrt{\frac{|S| \log 1/\delta}{N}}$$

$$\|Q^\pi - \hat{Q}^\pi\|_\infty = \|\gamma(I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) V^\pi\|_\infty$$

$$\leq \frac{\gamma}{1 - \gamma} \|(P - \hat{P}) V^\pi\|_\infty$$

$$\leq \frac{\gamma}{1 - \gamma} \left(\max_{s,a} \|P(\cdot|s, a) - \hat{P}(\cdot|s, a)\|_1 \right) \|V^\pi\|_\infty$$

$$\leq \frac{\gamma}{(1 - \gamma)^2} \left(\max_{s,a} \|P(\cdot|s, a) - \hat{P}(\cdot|s, a)\|_1 \right)$$

$$\leq \frac{\gamma}{(1 - \gamma)^2} c_2 \sqrt{\frac{|S| \log 1/\delta}{N}}$$

technically need to use the union bound here to account for probabilities

What does this mean?

$$\|Q^\pi - \hat{Q}^\pi\|_\infty \leq \frac{\gamma}{(1-\gamma)^2} c_2 \sqrt{\frac{|S| \log 1/\delta}{N}}$$

error grows **quadratically** in the horizon
each backup “accumulates” error

more samples = lower error

Some simple implications...

$$\|Q^\pi - \hat{Q}^\pi\|_\infty \leq \epsilon \qquad \epsilon = \frac{\gamma}{(1-\gamma)^2} c_2 \sqrt{\frac{|S| \log 1/\delta}{N}}$$

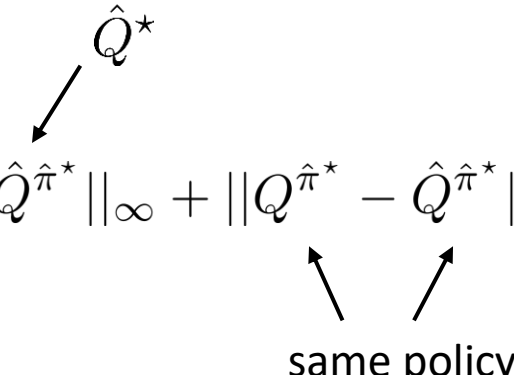
what about $\|Q^* - \hat{Q}^*\|_\infty$?

$$\left| \sup_x f(x) - \sup_x g(x) \right| \leq \sup_x |f(x) - g(x)|$$

$$\|Q^* - \hat{Q}^*\|_\infty = \left\| \sup_{\pi} Q^\pi - \sup_{\pi} \hat{Q}^\pi \right\|_\infty \leq \sup_{\pi} \|Q^\pi - \hat{Q}^\pi\|_\infty \leq \epsilon$$

what about $\|Q^* - Q^{\hat{\pi}^*}\|_\infty$?

$$\|Q^* - Q^{\hat{\pi}^*}\|_\infty = \|Q^* - \hat{Q}^{\hat{\pi}^*} + \hat{Q}^{\hat{\pi}^*} - Q^{\hat{\pi}^*}\|_\infty \leq \|Q^* - \hat{Q}^{\hat{\pi}^*}\|_\infty + \|Q^{\hat{\pi}^*} - \hat{Q}^{\hat{\pi}^*}\|_\infty \leq 2\epsilon$$



Part 4:

Analysis of model-free RL

Analyzing fitted Q-iteration

abstract model of exact Q-iteration:

$$Q_{k+1} \leftarrow TQ_k$$

Bellman operator
 $TQ = r + \gamma P \max_a Q$

abstract model of approximate fitted Q-iteration:

$$\hat{Q}_{k+1} \leftarrow \arg \min_{\hat{Q}} \|\hat{Q} - \hat{T}\hat{Q}_k\|$$

which norm?
 no convergence if $\|\cdot\|_2$
 we'll assume $\|\cdot\|_\infty$

Question: as $k \rightarrow \infty$, $\hat{Q}_k \rightarrow ?$

$$\lim_{k \rightarrow \infty} \|\hat{Q}_k - Q^*\|_\infty \leq ?$$

where do errors come from?

$$T \neq \hat{T} \quad \text{"sampling error"}$$

$$\hat{Q}_{k+1} \neq \hat{T}\hat{Q}_k \quad \text{"approximation error"}$$

approximate Bellman operator

$$\hat{T}Q = \hat{r} + \gamma \hat{P} \max_a Q$$

$$\hat{r}(s, a) = \frac{1}{N(s, a)} \sum_i \delta((s_i, a_i) = (s, a)) r_i$$

$$\hat{P}(s'|s, a) = \frac{N(s, a, s')}{N(s, a)}$$

Note: these are **not** models, this is the effect of averaging together transitions in the data!

Let's analyze sampling error

$$\hat{Q}_{k+1} \leftarrow \arg \min_{\hat{Q}} \|\hat{Q} - \hat{T}\hat{Q}_k\| \quad T \neq \hat{T}$$

vs. $T\hat{Q}_k$

$$\begin{aligned} |\hat{T}Q(s, a) - TQ(s, a)| &= |\hat{r}(s, a) - r(s, a) + \gamma(E_{\hat{P}(s'|s, a)}[\max_{a'} Q(s', a')] - E_{P(s'|s, a)}[\max_{a'} Q(s', a')])| \\ &\leq \underbrace{|\hat{r}(s, a) - r(s, a)|}_{\text{estimation error of continuous random variable}} + \gamma \underbrace{|(E_{\hat{P}(s'|s, a)}[\max_{a'} Q(s', a')] - E_{P(s'|s, a)}[\max_{a'} Q(s', a')])|}_{\text{just use Hoeffding's inequality directly!}} \end{aligned}$$

estimation error of continuous random variable
just use Hoeffding's inequality directly!

$$|\hat{r}(s, a) - r(s, a)| \leq 2R_{\max} \sqrt{\frac{\log 1/\delta}{2N}}$$

$$\begin{aligned} &\sum_{s'} (\hat{P}(s'|s, a) - P(s'|s, a)) \max_{a'} Q(s', a') \\ &\leq \sum_{s'} |\hat{P}(s'|s, a) - P(s'|s, a)| \max_{s', a'} Q(s', a') \\ &= \|\hat{P}(\cdot|s, a) - P(\cdot|s, a)\|_1 \|Q\|_{\infty} \\ &\leq c \|Q\|_{\infty} \sqrt{\frac{\log 1/\delta}{N}} \end{aligned}$$

Let's analyze sampling error

$$\hat{Q}_{k+1} \leftarrow \arg \min_{\hat{Q}} \|\hat{Q} - \hat{T}\hat{Q}_k\| \quad T \neq \hat{T}$$

vs. $T\hat{Q}_k$

$$|\hat{T}Q(s, a) - TQ(s, a)| \leq 2R_{\max} \sqrt{\frac{\log 1/\delta}{2N}} + c\|Q\|_{\infty} \sqrt{\frac{\log 1/\delta}{N}}$$

$$\|\hat{T}Q - TQ\|_{\infty} \leq 2R_{\max}c_1 \sqrt{\frac{\log |S||A|/\delta}{2N}} + c_2\|Q\|_{\infty} \sqrt{\frac{\log |S|/\delta}{N}}$$

using union bound

Let's analyze approximation error

approximation error assumption: $\|\hat{Q}_{k+1} - T\hat{Q}_k\|_\infty \leq \epsilon_k$

This is a strong assumption!

we'll analyze the exact backup operator for now,
but we'll come back to approximate backups later!

$$\begin{aligned}\|\hat{Q}_k - Q^*\|_\infty &= \|\hat{Q}_k - T\hat{Q}_{k-1} + T\hat{Q}_{k-1} - Q^*\|_\infty && \text{using fact that } Q^* \text{ is fixed point of } T \\ &= \|(\hat{Q}_k - T\hat{Q}_{k-1}) + (T\hat{Q}_{k-1} - TQ^*)\|_\infty \\ &\leq \|\hat{Q}_k - T\hat{Q}_{k-1}\|_\infty + \|T\hat{Q}_{k-1} - TQ^*\|_\infty \\ &\leq \epsilon_{k-1} + \|T\hat{Q}_{k-1} - TQ^*\|_\infty \\ &\leq \epsilon_{k-1} + \gamma\|\hat{Q}_{k-1} - Q^*\|_\infty && \text{using fact that } T \text{ is a } \gamma\text{-contraction}\end{aligned}$$

Let's analyze approximation error

$$\begin{aligned}\|\hat{Q}_k - Q^*\|_\infty &\leq \epsilon_{k-1} + \gamma \|\hat{Q}_{k-1} - Q^*\|_\infty \\ &\leq \epsilon_{k-1} + \gamma \epsilon_{k-2} + \gamma^2 \|\hat{Q}_{k-2} - Q^*\|_\infty \\ &\leq \epsilon_{k-1} + \gamma \epsilon_{k-2} + \gamma^2 \epsilon_{k-3} + \gamma^3 \|\hat{Q}_{k-3} - Q^*\|_\infty \\ &\leq \sum_{i=0}^{k-1} \gamma^i \epsilon_{k-i-1} + \gamma^k \|\hat{Q}_0 - Q^*\|_\infty\end{aligned}$$

$$\lim_{k \rightarrow \infty} \|\hat{Q}_k - Q^*\|_\infty \leq \sum_{i=0}^{\infty} \gamma^i \max_k \epsilon_k = \frac{1}{1-\gamma} \|\epsilon\|_\infty$$

approximation error scales with “horizon”

Putting it together

$$\|\hat{T}Q - TQ\|_\infty \leq 2R_{\max}c_1\sqrt{\frac{\log |S||A|/\delta}{2N}} + c_2\|Q\|_\infty\sqrt{\frac{\log |S|/\delta}{N}} \quad \text{“sampling error”}$$

$$\lim_{k \rightarrow \infty} \|\hat{Q}_k - Q^*\|_\infty \leq \frac{1}{1-\gamma} \max_k \epsilon_k = \frac{1}{1-\gamma} \max_k \|\hat{Q}_k - T\hat{Q}_{k-1}\|_\infty \quad \text{“approximation error”}$$

how much $\hat{Q}_{k+1}$ differs from $T\hat{Q}_k$
 due to: sampling error ($T \neq \hat{T}$)
 approximation error ($\hat{Q}_k \neq \hat{T}\hat{Q}_{k-1}$)

$$\begin{aligned} \|\hat{Q}_k - T\hat{Q}_{k-1}\|_\infty &= \|\hat{Q}_k - \hat{T}\hat{Q}_{k-1} + \hat{T}\hat{Q}_{k-1} - T\hat{Q}_{k-1}\|_\infty \\ &\leq \underbrace{\|\hat{Q}_k - \hat{T}\hat{Q}_{k-1}\|_\infty}_{\text{approximation}} + \underbrace{\|\hat{T}\hat{Q}_{k-1} - T\hat{Q}_{k-1}\|_\infty}_{\text{sampling}} \end{aligned}$$

What does it all mean?

$$\|\hat{T}Q - TQ\|_\infty \leq 2R_{\max}c_1\sqrt{\frac{\log|S||A|/\delta}{2N}} + c_2\|Q\|_\infty\sqrt{\frac{\log|S|/\delta}{N}} \quad \text{“sampling error”}$$

$\mathcal{O}\left(R_{\max}\frac{1}{1-\gamma}\right)$

$$\lim_{k \rightarrow \infty} \|\hat{Q}_k - Q^*\|_\infty \leq \frac{1}{1-\gamma} \max_k \epsilon_k = \frac{1}{1-\gamma} \max_k \|\hat{Q}_k - T\hat{Q}_{k-1}\|_\infty \quad \text{“approximation error”}$$

$$\|\hat{Q}_k - T\hat{Q}_{k-1}\|_\infty \leq \|\hat{Q}_k - \hat{T}\hat{Q}_{k-1}\|_\infty + \|\hat{T}\hat{Q}_{k-1} - T\hat{Q}_{k-1}\|_\infty$$

error “compounds” with horizon, over iterations and due to sampling

so far we needed strong (infinity norm) assumptions

more advanced results can be derived with p-norms under some distribution:

$$\|\hat{Q}_k - Q^*\|_{p,\mu} = \left(E_{(s,a) \sim \mu(s,a)} [\|\hat{Q}_k(s,a) - Q^*(s,a)\|^p] \right)^{1/p}$$